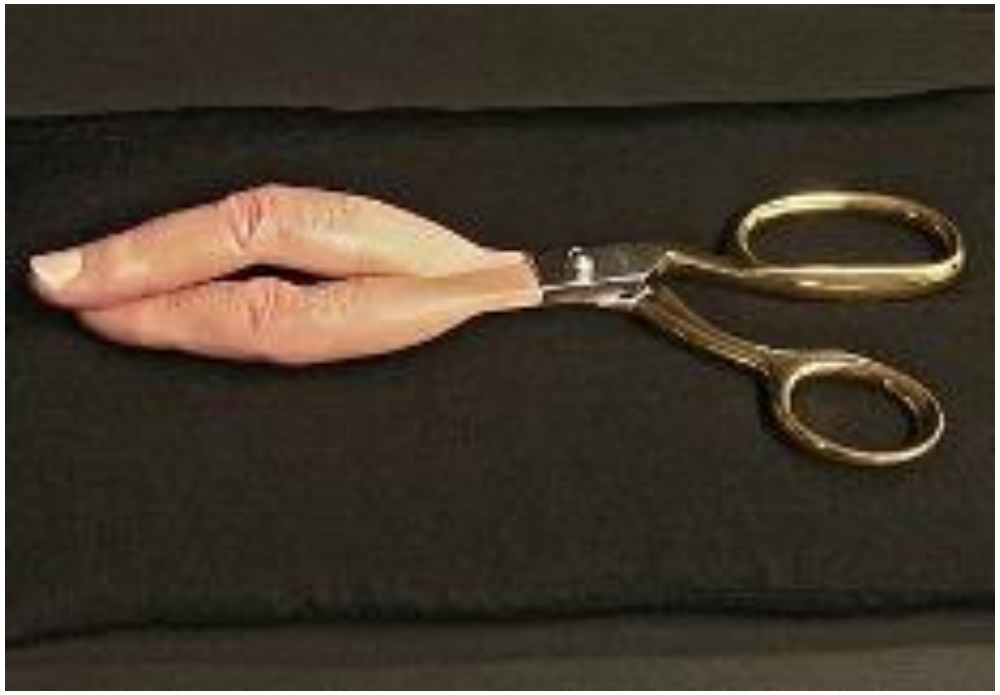
3 Marxian Perspectives

GEB2508 AI and Society

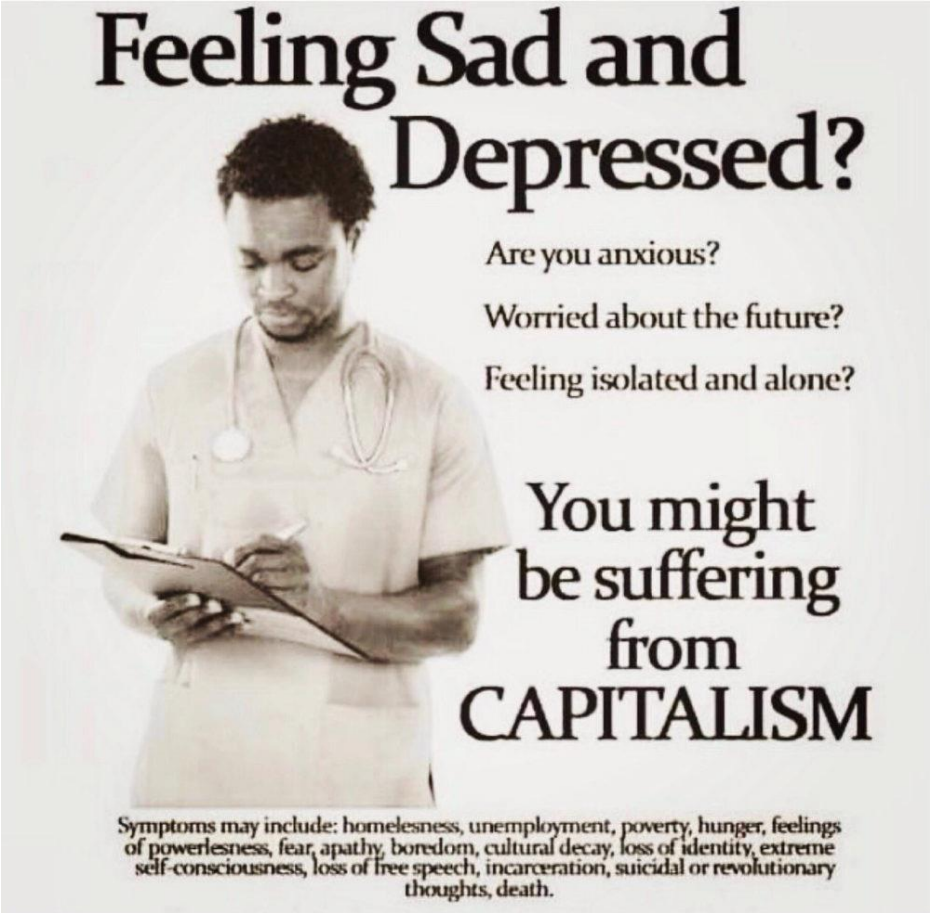
Review of Last Class

- Historiography of Science
  - Internalism vs. Externalism
  - The Sociological Turn
- Social Construction of Facts
  - SSK and The Strong Programme
  - Social Construction Theory
- Social Construction of Artifacts
  - Technological Determinism
  - Social Construction of Technology



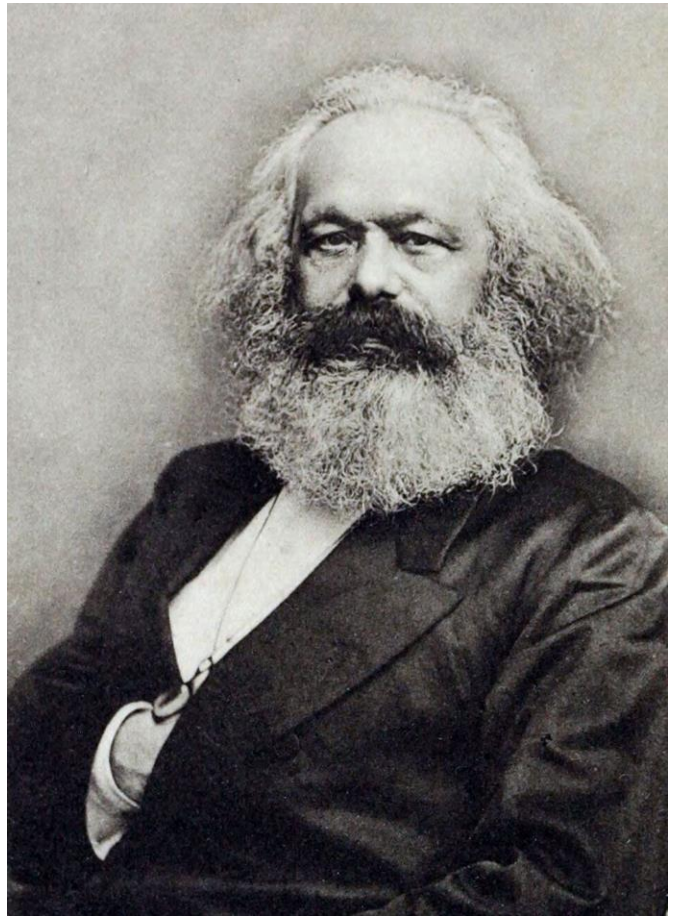
Content

- Marxist Theory
  - Labor and Value
  - Exploitation and Inequality
  - Base and Superstructure
- Marxian Perspectives on Technology
  - Machinery and Luddism
  - Division of Labor and Alienation
  - Technological Unemployment



Karl Marx (1818-1883)

- Born in Trier, Germany
- Studied law and philosophy at University of Bonn, University of Berlin, and University of Jena
- Moved to Paris and met Engels in 1844
- Lived in London for most of his Life (1949-1883)
- Typically cited as one of the principal architects of modern social science

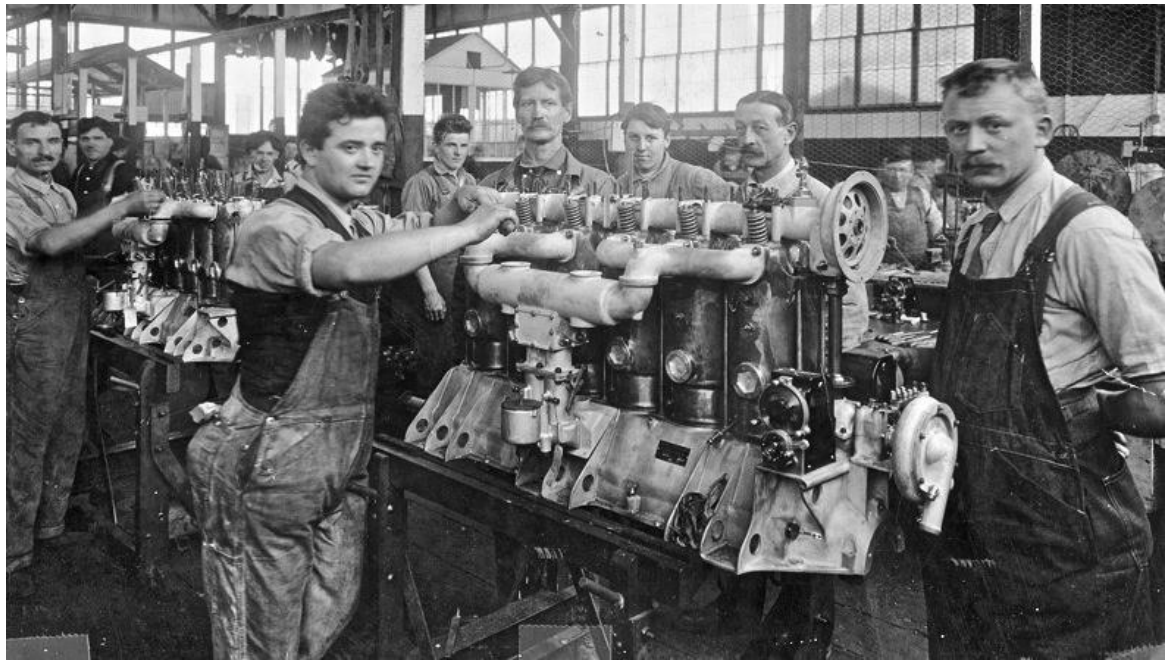


Marx Quotes

- History repeats itself, first as tragedy, second as farce.
- Capital comes dripping from head to foot, from every pore, with blood and dirt.
- Religion is the sigh of the oppressed creature, the heart of a heartless world, just as it is the spirit of a spiritless situation. It is the opium of the people.
- There is a specter haunting Europe, the specter of Communism.
- Workers of the world unite; you have nothing to lose but your chains.
- The philosophers have only interpreted the world, in various ways. The point, however, is to change it.

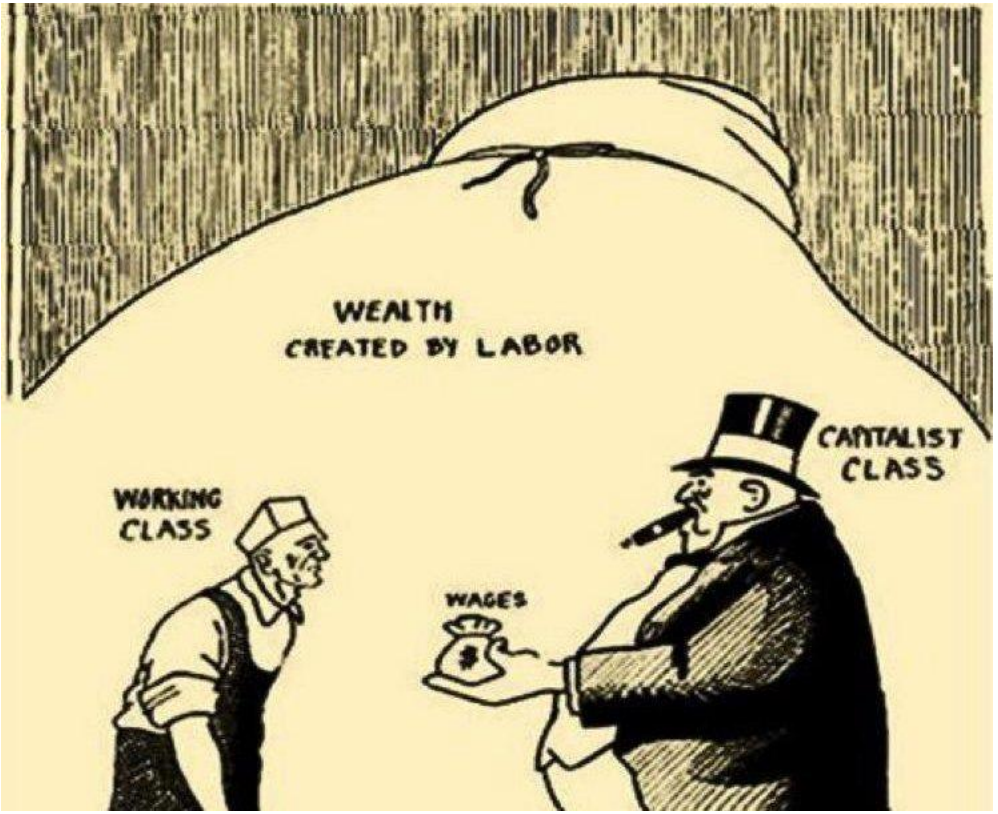
Historical Background (19C)

- Industrial Revolution
  - Britain → European continent and North America
- Men, women, and children in factories
- No welfare state
- Long hours, low pay
- Periodic unemployment
- Protests and movements



Class

- A class is those who share common **economic** interests and engage in collective actions to advance their interests.
- Marx defines social classes in terms of one's relationship to the **means of production**.
- Capitalism causes a **conflict of interest** between the ruling class (the bourgeoisie) and the working class (the proletariat).

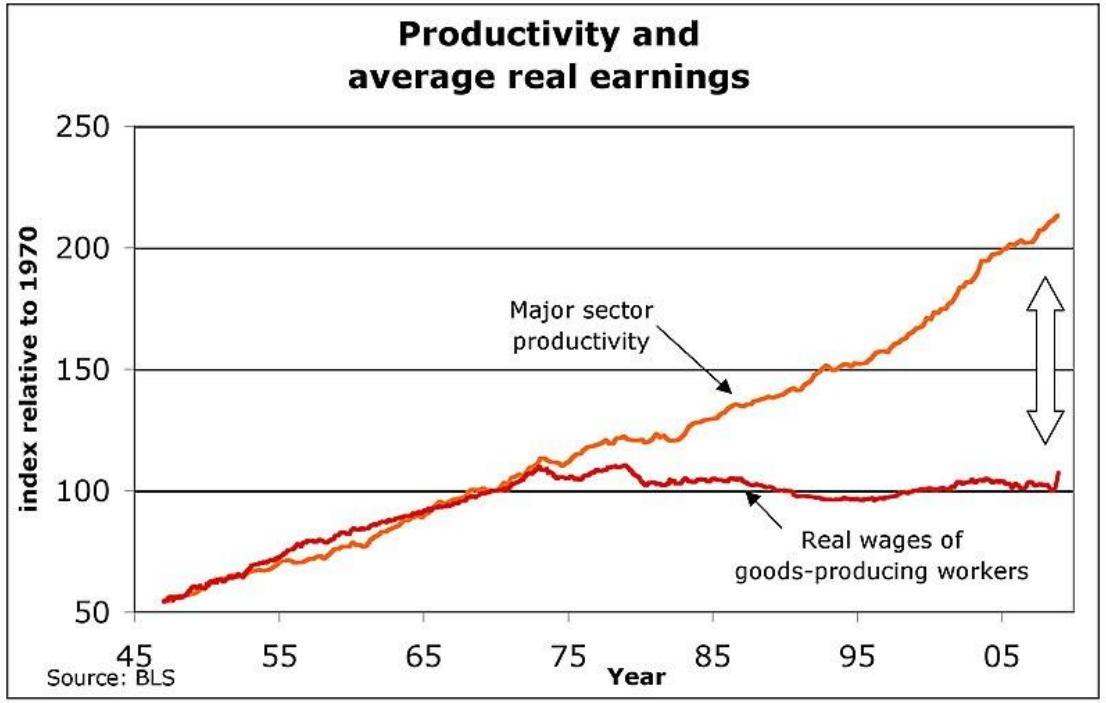


Exploitation

- **Surplus labor**: labor performed more than the labor **necessary** to produce the means of livelihood of the worker.
- **Surplus value**: the surplus produced over and above what is required to survive, which is translated into **profit** in capitalism.

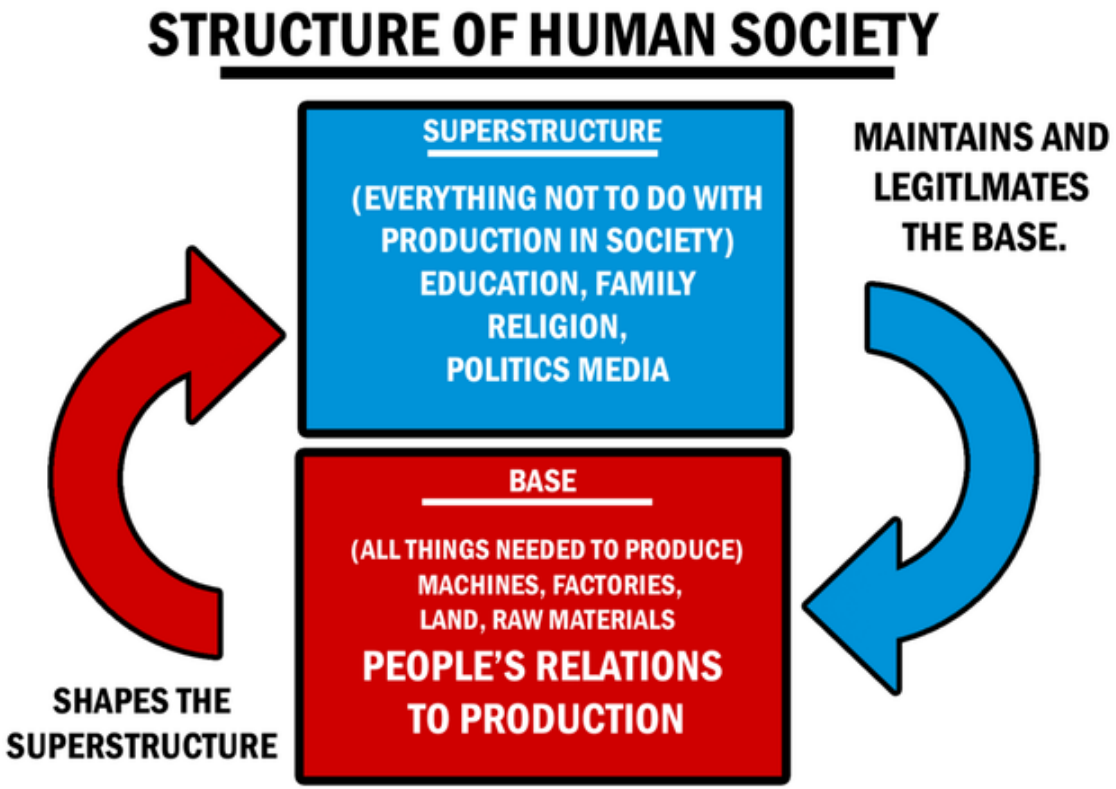
Inequality

- Capitalism promotes private accumulation of wealth and advocates private property
- **Unequal distribution** of income and wealth
  - The globe's billionaires are wealthier than half the planet.
- New technologies, including AI, are exacerbating existing wealth disparities.



The Base and the Superstructure

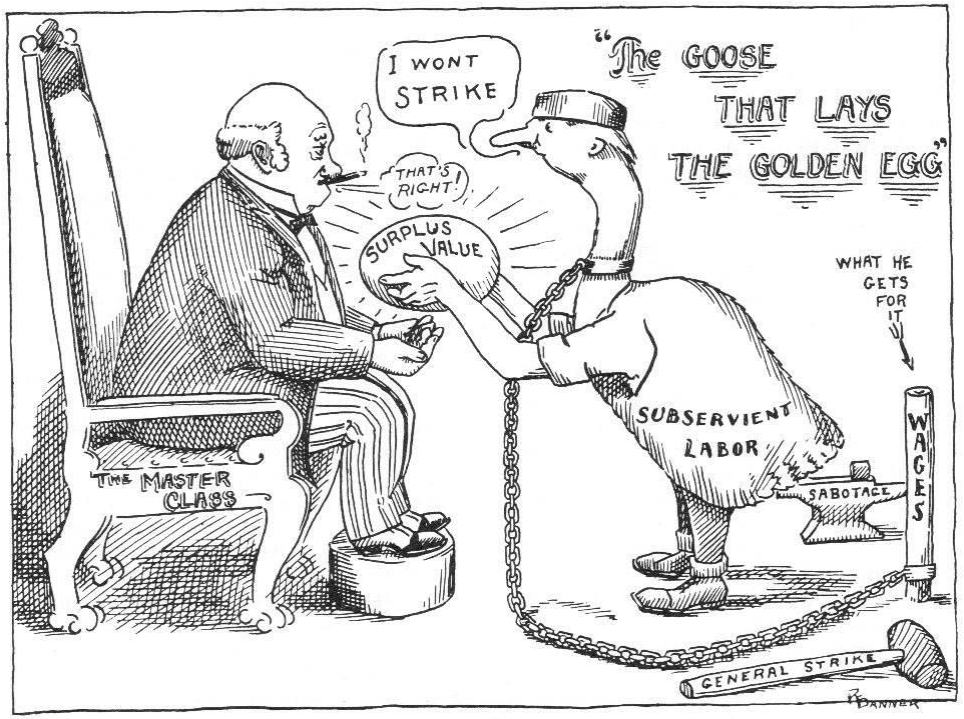
- The Base
  - **Economic** in nature; the base determines the superstructure
- The Superstructure
  - To control the proletariat into conforming to the beliefs, ideas, and values that **keep** the bourgeoisie in power
  - State apparatus and ideological apparatus



False Consciousness

- Marx believed the proletariat existed in a state of false class consciousness – they unwittingly **misperceive** their real position in society and systematically **misunderstand** their genuine interests.
  - E.g., education, media, family
- There is no escape from ideology. It is the "prison-house of language." (Fredrick Jameson)

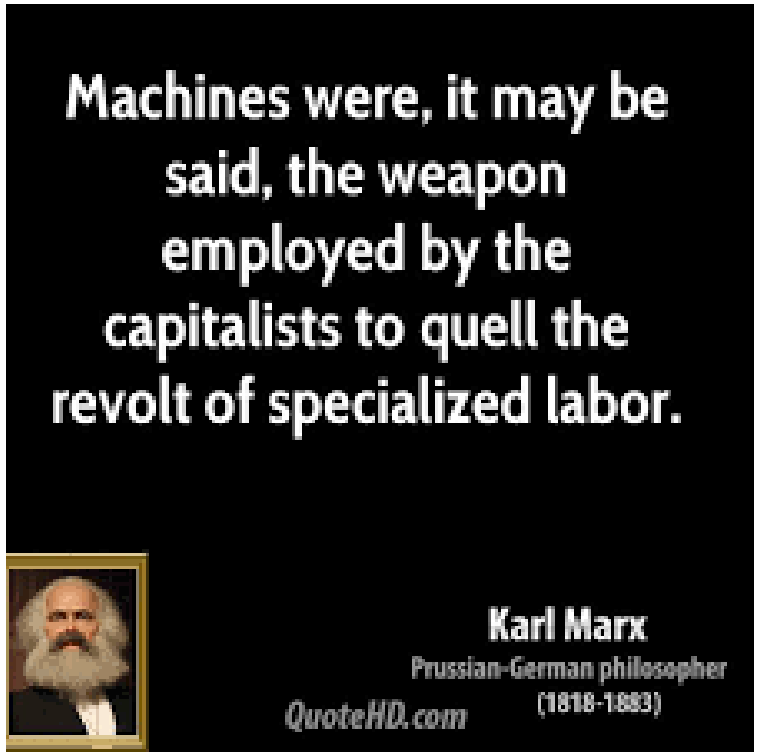
Industrial Worker, February 13, 1913.



WHY SHOULD WORKERS PRODUCE FOR IDLERS?

Marx on Machinery

- “To write a whole history of the inventions made since 1830 for the sole purpose of providing capital with **weapons against** working-class revolt.”
- “The most powerful instrument for **reducing** labour-time suffers a dialectical inversion and becomes the most unfailing means for turning the **whole lifetime** of the worker into labour-time at capital’s disposal for its own valorisation.”
- “The laborer becomes a mere **appendage** to an already existing material condition of production.”



Does Machinery Liberate or Enslave Us?

Economist in 1930: A hundred years from now, humans will only need to work 3 hours a day.

Worker in 2030: I can't be replaced by robots, so I have to work 16 hours a day.

Luddite Movement

- Ned Ludd was an apprentice who smashed two machines with a hammer
- The Luddites: an organization of workers who smashed newly developed **labor-saving machinery** (1811-1817)
- Frame Breaking Act (1812) made industrial sabotage a capital crime.

Modern Luddites

美团也在抓紧研制的无人配送，有一个细节，前阵子和一个曾经在美团的朋友聊天，她说，他们的无人配送车，遇到的一个问题是，经常被路过的骑手踹一脚。

A former Meituan employee mentioned that one of the issues their unmanned delivery vehicles often encounter is being kicked by passing delivery riders.

Living Labor vs. Dead Labor

- Living labor: human labor; only source of new value
- Dead labor: past labor embodied in **means of production**; transfers (not creates) new value
  - Classical forms: machinery, tools, factory, capital
  - Today: algorithms, models, datasets, platforms
- Dead labor **dominates** living labor, as capitalists always seek to replace living labor with dead labor
- AI as dead labor
  - AI systems embody massive dead labor
  - Control of dead labor → concentration of power

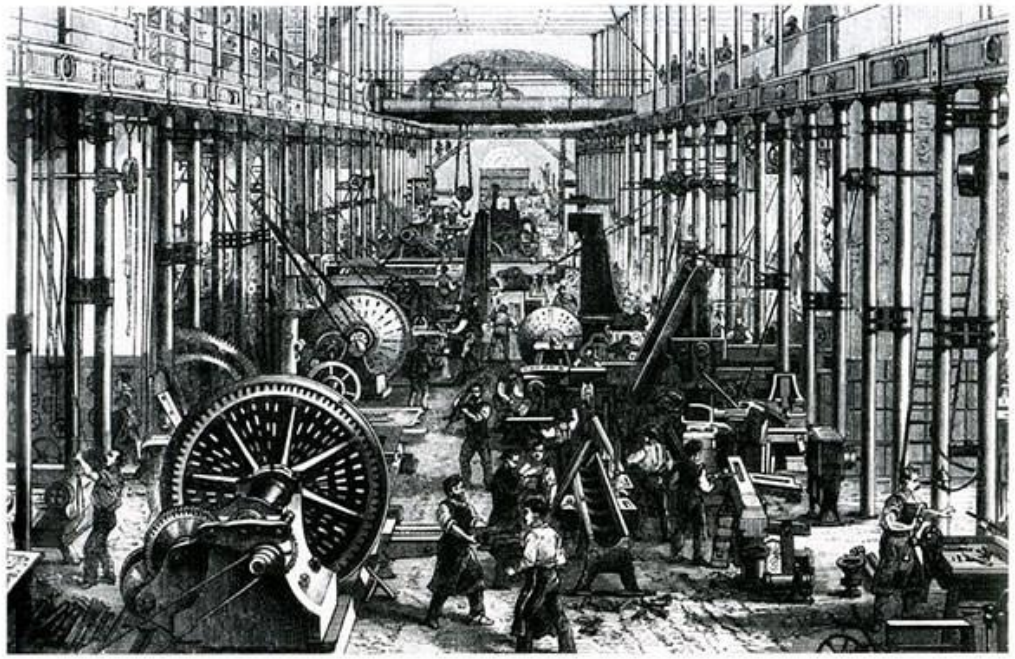
“ Capital is dead labour, that, vampire-like, only lives by sucking living labour, and lives the more, the more labour it sucks. ”

Karl Marx

(Das Kapital)

Factory System

- The factory system is a method of manufacturing using **machinery** and division of labor.
  - Fusion of capital, science, and technical knowledge
- From traditional artisan craftsmanship to the factory system
  - Reduced the demand for **skill**, making workers easily **replaceable**.
  - Workers lost control over their means of production and labor.



Taylorism & Fordism

- Taylorism: the assembly line and scientific management
- Ford's assembly line: the system of **mass** production through **standardized** products
  - A Ford worker: "The machine I'm on goes at such a terrific speed that I can't help stepping on it to keep up with the machine. It's my boss."

Diego Rivera, Detroit Industry Murals, 1932

Division of Labor and Alienation

- Adam Smith on division of labor: breaking production into tasks increases **efficiency**; specialization fosters new **inventions** that make tasks easier.
- Marx views the division of labor as central to capitalist exploitation
- Alienation: the feeling of being **separated** from one's human nature and from the products of one's labor; a sense of **disconnection** and estrangement

一顆螺絲掉在地上

一顆螺絲掉在地上
在這個加班的夜晚
垂直降落, 輕輕一響
不會引起任何人的注意

就像在此之前
某個相同的夜晚
有個人掉在地上

——許立志(富士康工人, 24歲時墜樓自殺)

Modern Times (1936)

- How does Modern Times portray the relations between machinery and labor?
- How does Modern Times illustrate Marx's theory of exploitation and alienation?

Taylorism through AI

- Micro-tasks and platform work
  - **Data** assembly line
- Algorithmic management
  - Dashboards, productivity metrics, real-time monitoring
  - Decision-making embedded in algorithms and code.
- **Invisible** human labor
  - Data labeling, model testing, content moderation, etc.

开会学到一个新词：数据采集厂。现在各地出现了这样的工厂，雇了很多人在不停重复地叠衣服、拧螺丝、叠衣服、拧螺丝……为了通过动捕设备采集大量人类运动轨迹，用来训练人形机器人。

Data collection factories are now emerging in which large numbers of workers repeatedly perform routine tasks - such as folding clothes or tightening screws - to generate extensive human motion trajectories via motion-capture systems. These data are subsequently used to train humanoid robots.

The Illusion of Full Autonomy

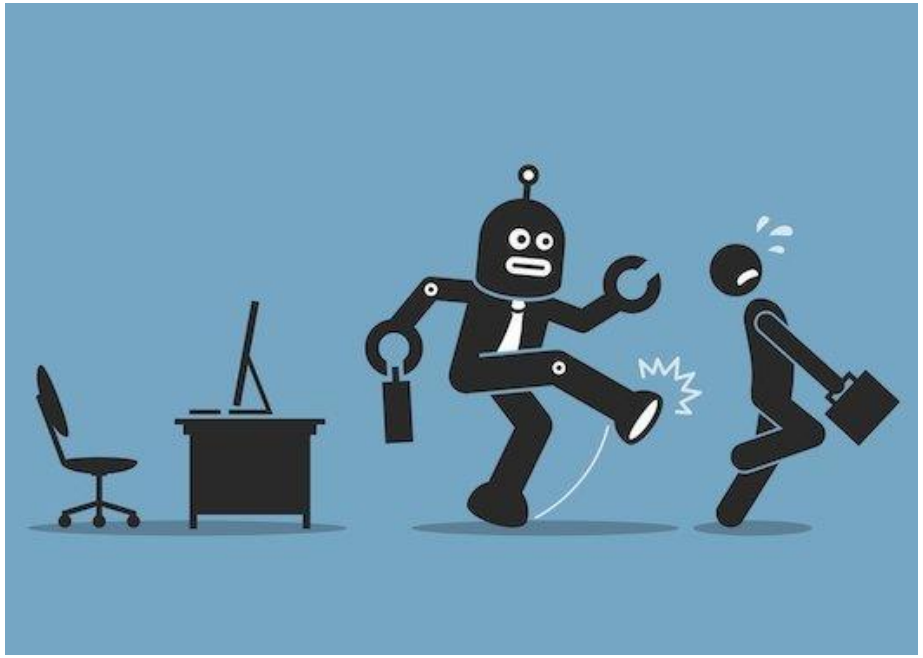
- Potemkin AI: systems presented as autonomous actually rely heavily on **hidden** human labor
- Claim: fully autonomous systems
- Reality: hidden human labor
  - Driverless vehicles with safety drivers
  - Content moderation presented as AI filtering

Structural Unemployment

- The **reserve army of labor** refers to the presence of a large pool of workers living in precarious conditions
- **Oversupply** of labor is **continually** produced under capitalism. Its very existence serves to **discipline** and exploit the proletariat.
- This ‘surplus population’ is **relative**, not **absolute**.

Technological Unemployment

- As technology advances, some forms of work become obsolete, leading to technological unemployment.
- Will technological unemployment result in long-term unemployment?
  - John Maynard Keynes: only a **temporary** phase of maladjustment
  - Marx: technological unemployment is **structural** for sustaining the reserve army of labor.



Platform-based Gig Work

- Wage instability and low pay
  - Unpredictable earnings
- Lack of transparency and bargaining power
  - Information **asymmetry**
- Precarity and inequality
  - Risks **shifted** onto workers; undermined labor protection
- Algorithmic control

r/confession · 3h ago
Trowaway_whistleblow

I'm a developer for a major food delivery app. The 'Priority Fee' and 'Driver Benefit Fee' go 100% to the company. The driver sees $0 of it.

I'm posting this from a library Wi-Fi on a burner laptop because I am technically under a massive NDA. I don't care anymore. I put in my two weeks yesterday and honestly, I hope they sue me. I've been sitting on this for about eight months, just watching the code getting pushed to production, and I can't sleep at night knowing I helped build this machine.

You guys always suspect the algorithms are rigged against you, but the reality is actually so much more depressing than the conspiracy theories. I'm a backend engineer. I sit in the weekly sprint planning meetings where Product Managers (PMs) discuss how to squeeze another 0.4% margin out of "human assets" (that's literally what they call drivers in the database schemas). They talk about these people like they are resource nodes in a video game, not fathers and mothers trying to pay rent.

First off, the "Priority Delivery" is a total scam. It was pitched to us as a "psychological value add." Like I said in the title, when you pay that extra $2.99, it changes a boolean flag in the order JSON, but the dispatch logic literally ignores it. It does nothing to speed you up.

We actually ran an A/B test last year where we didn't speed up the priority orders, we just purposefully delayed non-priority orders by 5 to 10 minutes to make the Priority ones "feel" faster by comparison. Management loved the results. We generated millions in pure profit just by making the standard service worse, not by making the premium service better.

But the thing that actually makes me sick—and the main reason I'm quitting—is the "Desperation Score." We have a hidden metric for drivers that tracks how desperate they are for cash based on their acceptance behavior.

If a driver usually logs on at 10 PM and accepts every garbage $3 order instantly without hesitation, the algo tags them as "High Desperation." Once they are tagged, the system then deliberately stops showing them high-paying orders. The logic is: "Why pay this guy $15 for a run when we know he's desperate enough to do it for $6?" We save the good tips for the "casual" drivers to hook them in and gamify their experience, while the full-timers get grinded into dust.

Then there is the "Benefit Fee." You've probably seen that $1.50 "Regulatory Response Fee" or "Driver Benefits Fee" that appeared on your bill after the recent labor laws passed. The wording is designed to make you feel like you're helping the worker.

In reality, that money goes straight to a corporate slush fund used to lobby against driver unions. We have a specific internal cost center for "Policy Defense," and that fee feeds directly into it. You are literally paying for the high-end lawyers that are fighting to keep your delivery guy homeless.

And regarding tips, we're essentially doing Tip Theft 2.0. We don't "steal" them legally anymore because we got sued for that. Instead, we use predictive modeling to dynamically lower the base pay.

If the algo predicts you are a "high tipper" and you'll likely drop $10, it offers the driver a measly $2 base pay. If you tip $0, it offers them $8 base pay just to get the food moved. The result is that your generosity isn't rewarding the driver; it's subsidizing us. You're paying their wage so we don't have to.

I'm drunk and I'm angry. Ask me anything before this gets taken down.

12K 671 18 Share

了解到一组数据，全国快递员、滴滴司机和外卖小哥大概有将近一个亿，其实在茫然不自知的情况下，已经至少有超过一个亿的人工作在算法设定的模型里，有段时间我经常打车，和多个司机聊天，基本上证实了一个事情，无论司机每天多么努力，每小时平均收入是无限趋近于30块钱的，当你运气好接到一个长途大单后，返程往往接不到客，当你前面几个小时收入不错的时候，后面几个小时直线下降，算法可以用回归和拟合控制收入曲线，你可以跑得赢红绿灯，但绝跑不赢算法，滴滴如此，外卖和跑腿亦是如此。人类为隐藏在数据中心的算法大脑打工的剧情已经悄然实现，只是很多人没有意识到而已。

11.1k I'm a developer for a major food delivery app. The 'Priority Fee' and 'Driver Benefit Fee' go 100% to the company. The driver sees $0 of it. (self.confession)
submitted 3 hours ago by Trowaway_whistleblow

我是某家大型外卖平台的后台工程师，今天就把这些事情说清楚吧！你在账单里看到的优先配送费和司机福利费，其实司机完全拿不到，全部都进公司口袋。我看着这些机制，越想越觉得心里过不去。

优先配送根本不是真的更快，用户付钱之后，系统只是把订单上改了个无关紧要的小标记。调度不会因为你多付几块就帮你加速，我们之前做过实验，不是让付费的更快，而是故意让普通订单更慢几分钟。这样付费的人就会以为自己买到了加速服务。公司靠这个赚了好几百万。

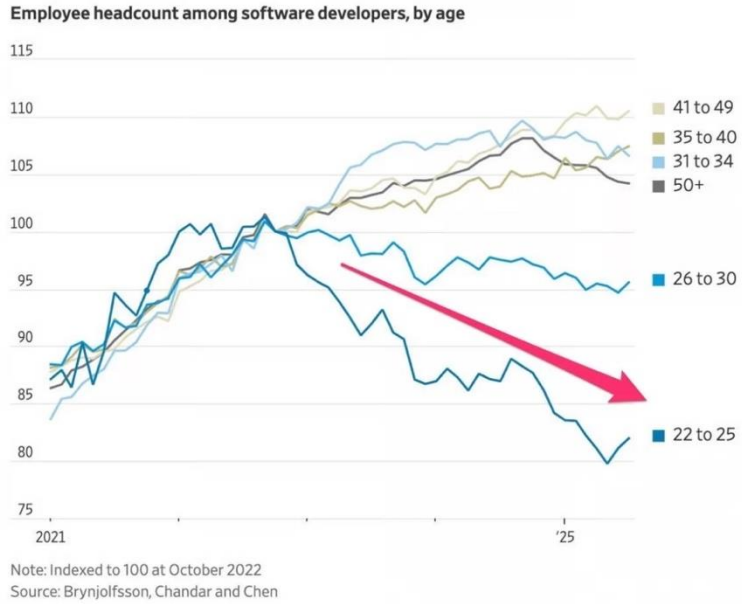
内部还有一个让我最恶心的指标叫绝望分。系统会根据司机的行为判断他是不是特别缺钱。比如深夜上线、低价订单立即抢这种情况。只要被判定成很缺钱，系统就不会给这些司机推高价单。逻辑非常简单：既然他几美元的单也肯跑，那没必要把好的单给他，我们把比较赚钱的订单留给偶尔上线的人，让他们觉得体验不错、愿意继续玩，而真正靠跑单生活的司机就很难受。

账单里的司机福利费听起来像是帮司机争取权益。实际上钱全部被公司拿去请律师、做公关、对抗司机工会。用户以为自己在支持司机。其实是在帮助公司压住司机的声音。内部有专门的成本中心来负责这件事。钱从来没真正落到司机手里。

小费方面，公司不再直接动手脚因为之前吃过官司。但换了一种更隐蔽的方式。系统会根据用户平常给小费的习惯来调整司机的基础收入。越可能给高额小费的用户，系统给那单司机的基础收入就越低。结果就是用户的小费越多，公司承担的成本越少。司机并没有真正因为你的慷慨而赚到更多。

整个系统就是这样设计的。让用户以为自己支付的是更好的服务。让司机以为努力多跑就会赚得更多。实际上你付的额外费用和司机付出的辛苦都先被公司拿走。想想真的很荒唐。

AB KUAIDONG

AI and Job Displacement

- AI shifts automation from **physical** to **cognitive** and **affective** labor
  - Occupations most exposed: **routine** cognitive and manual work
  - Labor markets **polarization**: high-skill and low-skill growth
- Automation as a key driver of increasing **inequality**